Classifying Skin Cancer Using Classical Computer Vision

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Current Questions

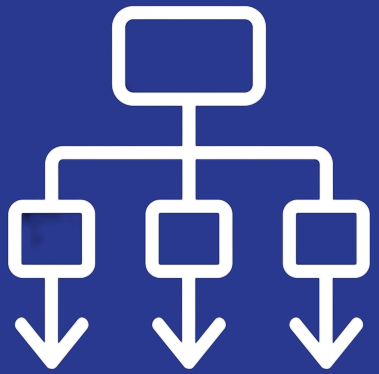
Feature Importance → Feature Distributions → Hair Removal

Which features contributed the most/least to the models?

How are feature values distributed across the classes?

How does hair removal affect feature distribution?

XG Boost



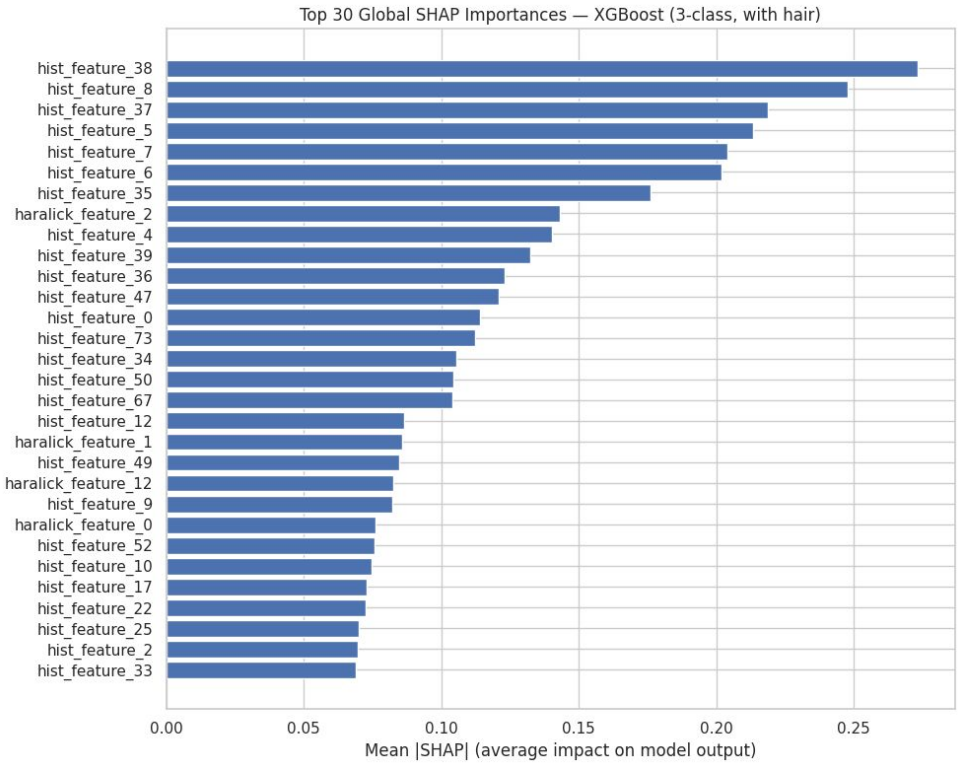
Feature Importance: Permutation Importance

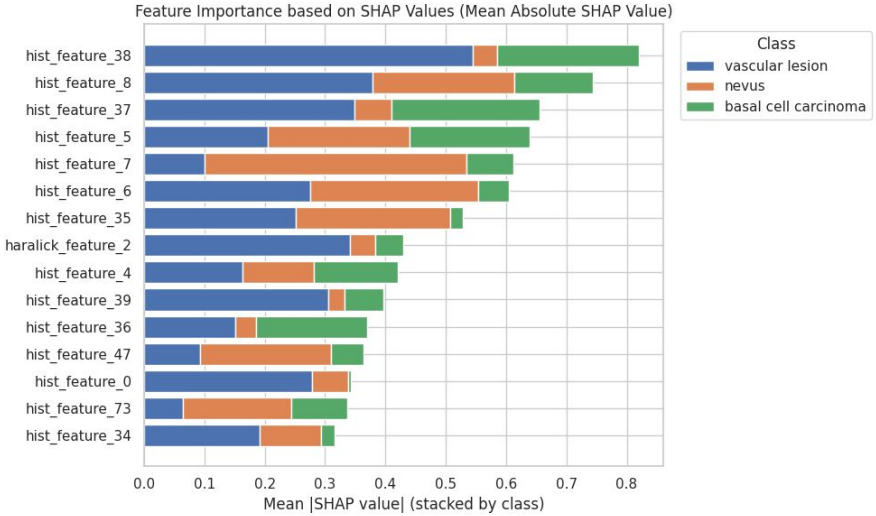
Quick sanity metrics (same split you're using for PI):					
Accuracy:	0.8542				
		precesion	recall	f1-score	support
vascular lesion		1	0.69	0.81	16
nevus		0.89	1	0.94	16
basal cell carcinoma		0.74	0.88	0.8	16
	accuracy			0.85	48
	macro avg	0.88	0.85	0.85	48
	weighted avg	0.88	0.85	0.85	48

Bottom 10 features by permutation importance:

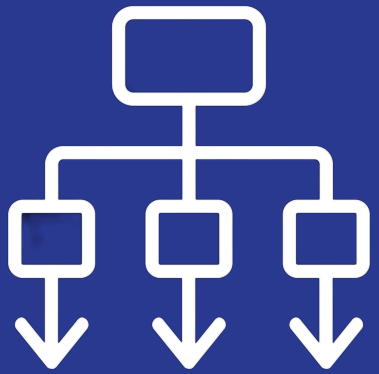
	feature	pi_mean	pi_std
42	hist_feature_39	-0.019608	0.018038
110	haralick_feature_11	-0.019757	0.006586
4	hist_feature_1	-0.021170	0.013501
100	haralick_feature_1	-0.021304	0.013501
8	hist_feature_5	-0.021704	0.016624
41	hist_feature_38	-0.021768	0.019278
111	haralick_feature_12	-0.021903	0.009707
11	hist_feature_8	-0.023343	0.022325
97	hist_feature_94	-0.027508	0.013770
39	hist_feature_36	-0.032019	0.022215

Feature Importance: SHAP

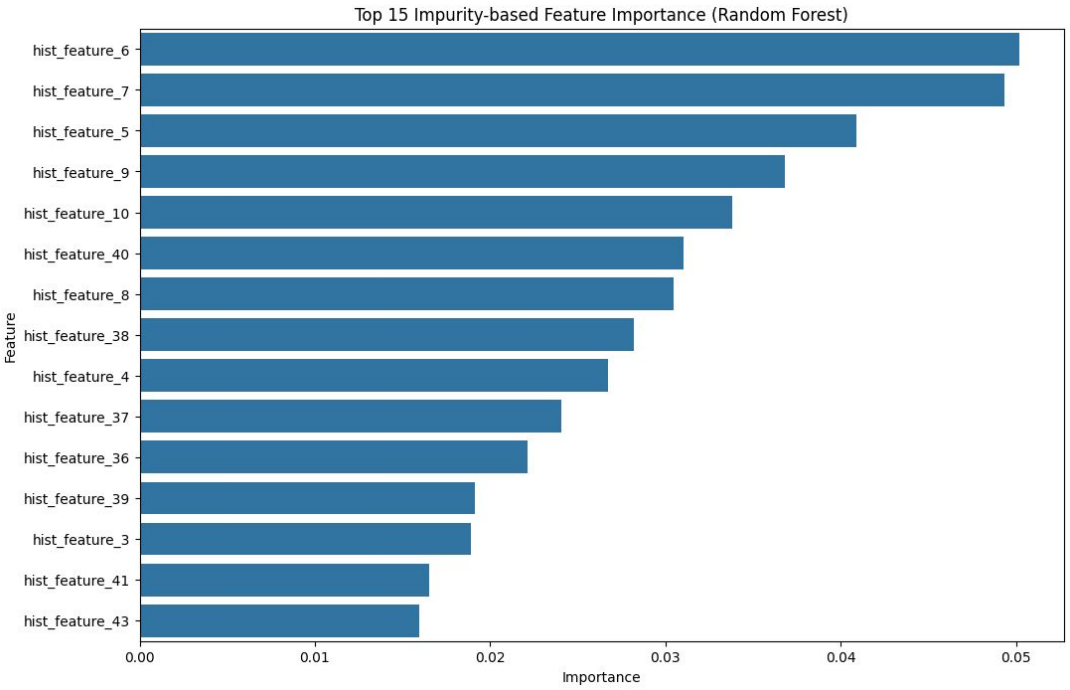


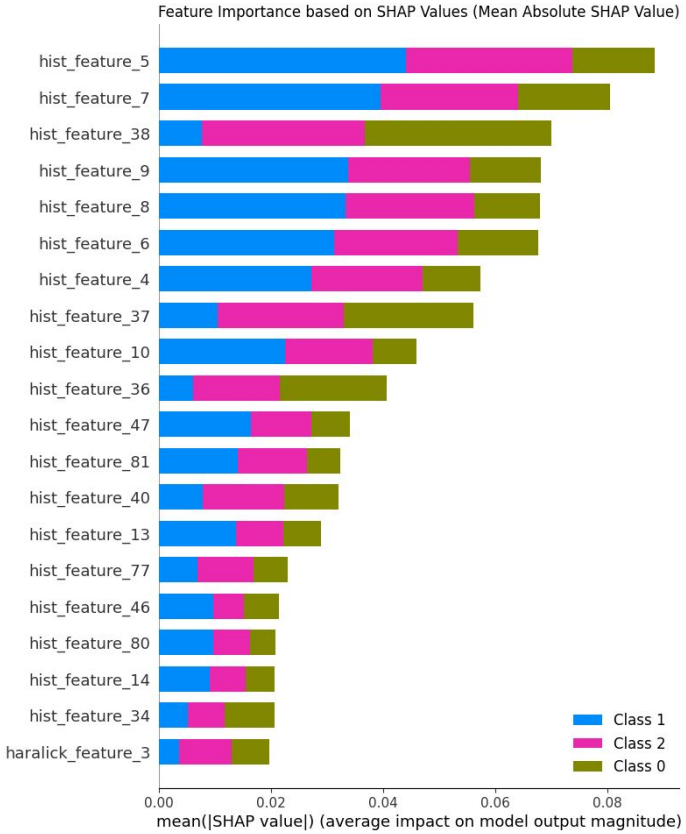


Random Forest



Feature Importance





Feature Premutation

```
[ 0.          0.          0.          0.          0.          0.
 -0.00875    -0.005      -0.005      -0.02666667 -0.01916667  0.
 -0.00541667  0.         -0.0075      0.          0.          0.
  0.          0.          0.          0.          0.          0.
  0.          0.          0.          0.          0.          0.
  0.          0.          0.          0.          0.          0.
  0.         -0.00958333 -0.01166667 -0.04416667 -0.04333333 -0.05166667
 -0.02583333 -0.01875     0.          0.          0.          0.
  0.          0.          0.          0.          0.          0.
  0.          0.          0.          0.          0.          0.
  0.          0.          0.          0.          0.          0.
  0.          0.          0.          0.          0.          0.
  0.          0.          0.          0.          0.          0.
 -0.00541667  0.          0.          0.          0.          0.
  0.          0.          0.          0.          0.          0.
  0.          0.          0.          0.          0.          0.
  0.          0.          0.          0.          0.         -0.00458333
 -0.00958333  0.          0.          0.          0.          0.
  0.          0.          0.          0.        ]
```

- Importance is always near zero
- Some features increased accuracy by a little and most features are redundant
- Highly correlated features
- Features compensate for one another

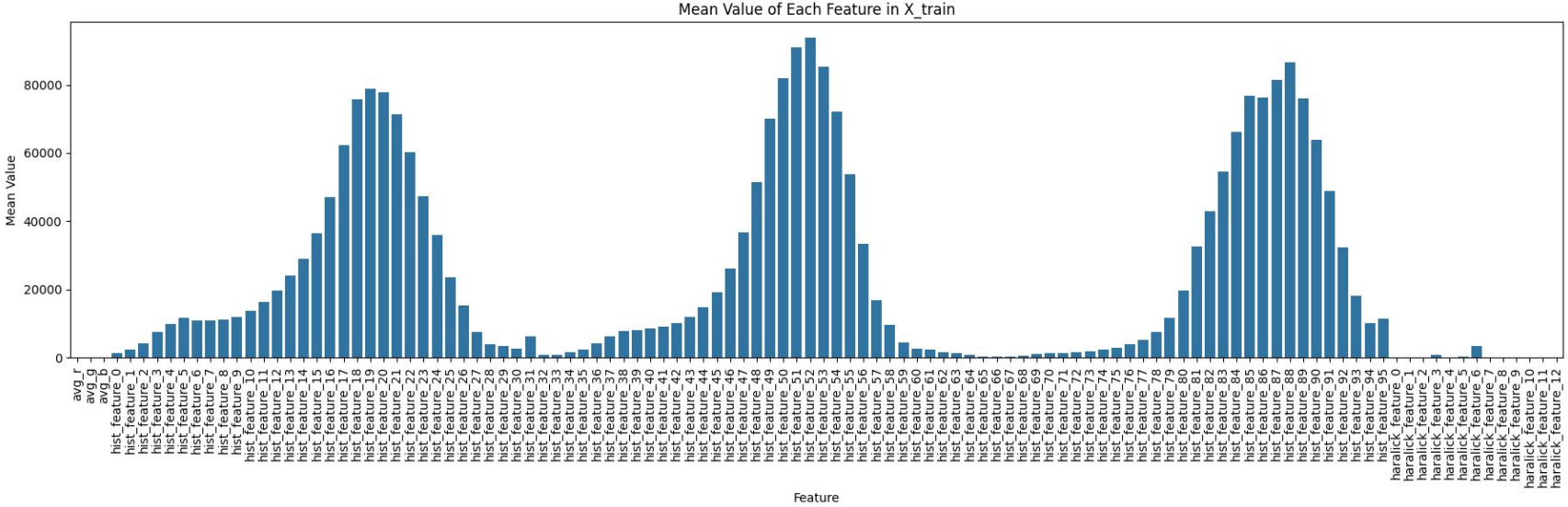
Feature Premutation Cont.

```
[0.         0.         0.         0.         0.         0.
 0.01109461 0.00889757 0.00889757 0.01818119 0.01493039 0.
 0.00913821 0.         0.01       0.         0.         0.
 0.         0.         0.         0.         0.         0.
 0.         0.         0.         0.         0.         0.
 0.         0.         0.         0.         0.         0.
 0.         0.01193879 0.01391941 0.02109831 0.01658312 0.02051422
 0.0147667  0.0151669  0.         0.         0.         0.
 0.         0.         0.         0.         0.         0.
 0.         0.         0.         0.         0.         0.
 0.         0.         0.         0.         0.         0.
 0.         0.         0.         0.         0.         0.
 0.         0.         0.         0.         0.         0.
 0.00913821 0.         0.         0.         0.         0.
 0.         0.         0.         0.         0.         0.
 0.         0.         0.         0.         0.         0.
 0.         0.         0.         0.         0.         0.00863013
 0.01038328 0.         0.         0.         0.         0.
 0.         0.         0.         0.        ]
```

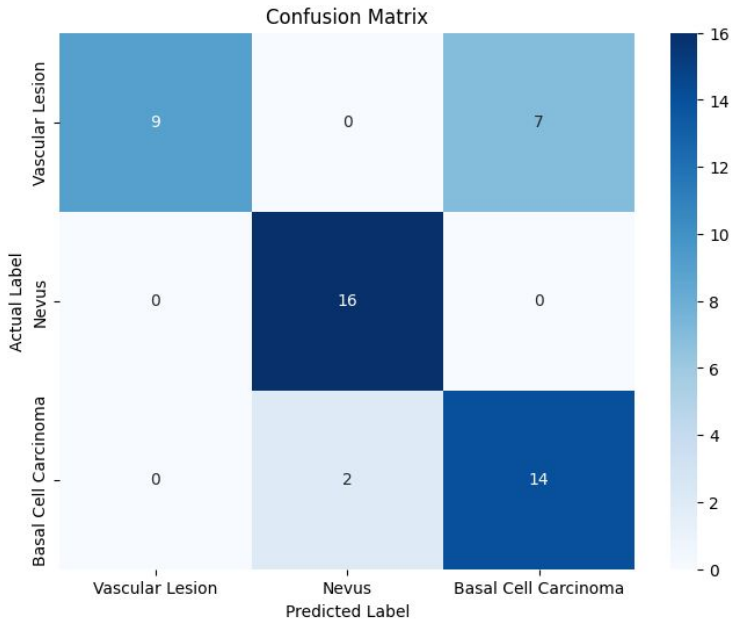
- Stability of the features contributions
- Features are weak predictors
- No single dominant features
- 0.008 ~ 0.02

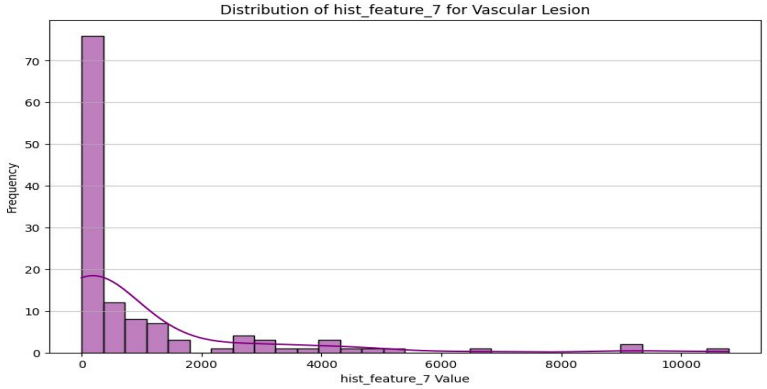
Feature Analysis

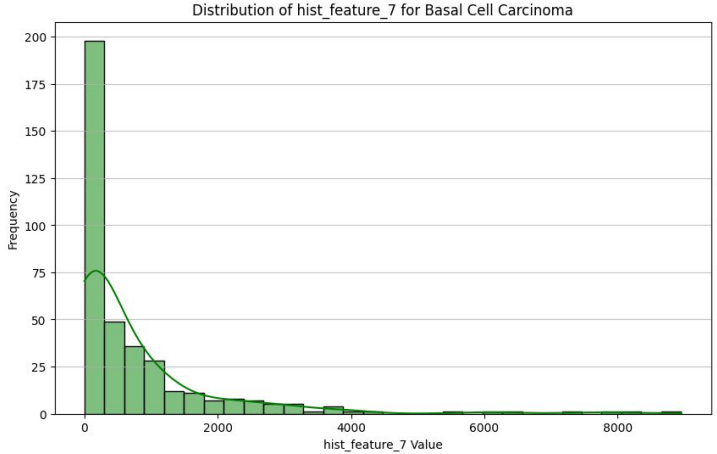
- Histogram Features
  - 3 Channels RGB - (0-32 Red, 33-64 Green, 65-96 Blue)
  - Color Intensity ranges 0-225
- Each value indicates how frequently pixel brightness values appear in that intensity range.



Feature Analysis Cont.







Metrics

Baseline

Classification Report:

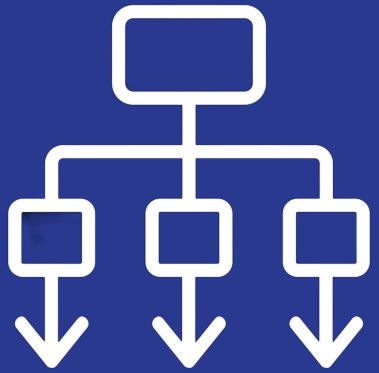
	precision	recall	f1-score	support
0	1.00	0.56	0.72	16
1	0.89	1.00	0.94	16
2	0.67	0.88	0.76	16
accuracy			0.81	48
macro avg	0.85	0.81	0.81	48
weighted avg	0.85	0.81	0.81	48

Reduced Features

Classification Report:

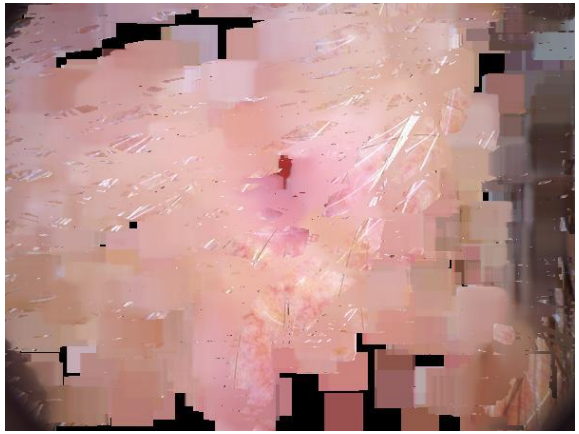
	precision	recall	f1-score	support
0	1.00	0.62	0.77	16
1	0.89	1.00	0.94	16
2	0.70	0.88	0.78	16
accuracy			0.83	48
macro avg	0.86	0.83	0.83	48
weighted avg	0.86	0.83	0.83	48

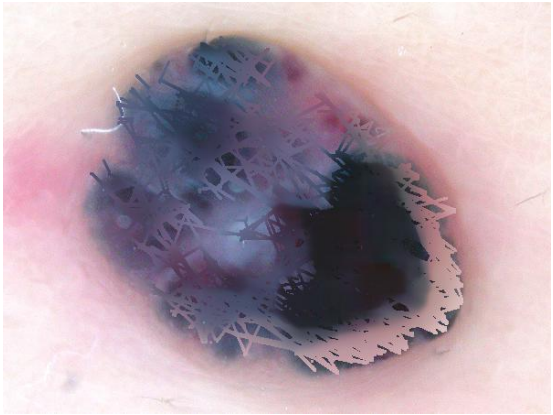
Preprocessing and Feature Distributions



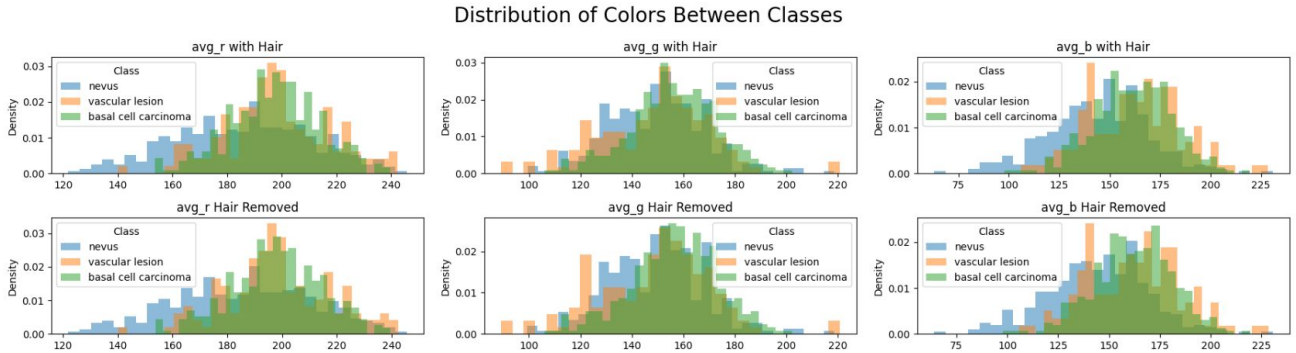
Original Processing Pipeline

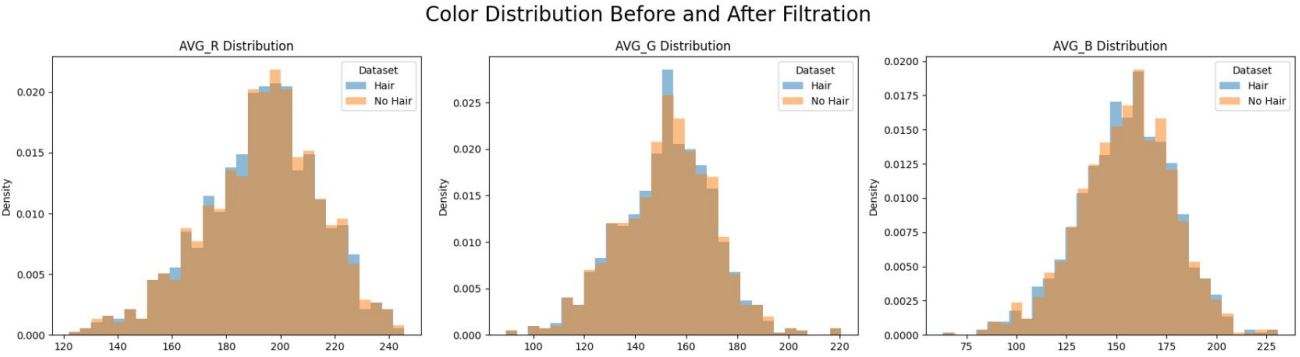
- Algorithm based off of open source Kaggle function, deletes detected hair pixels
- Led to several cases of ‘corrupted’ images
- Pipeline took 1.5 hours to complete

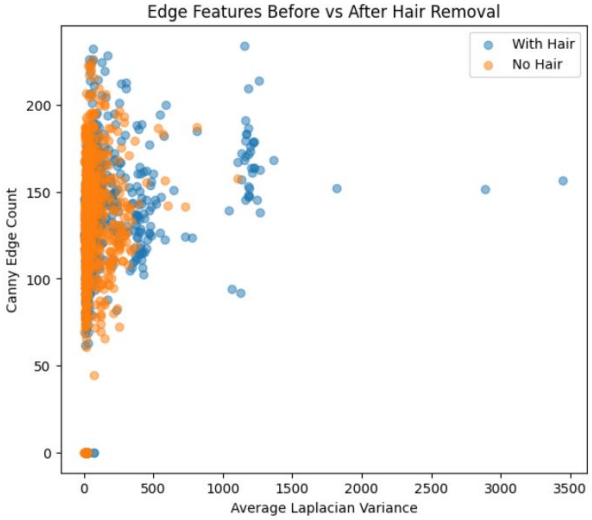




Analysis of Previous Dataset

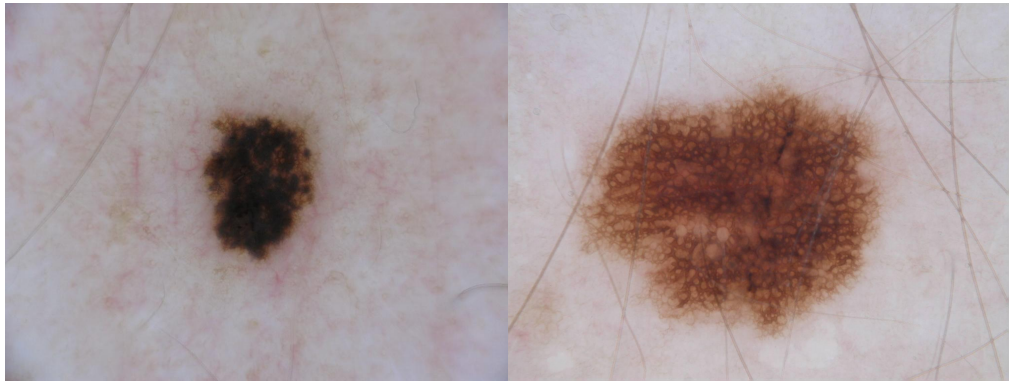




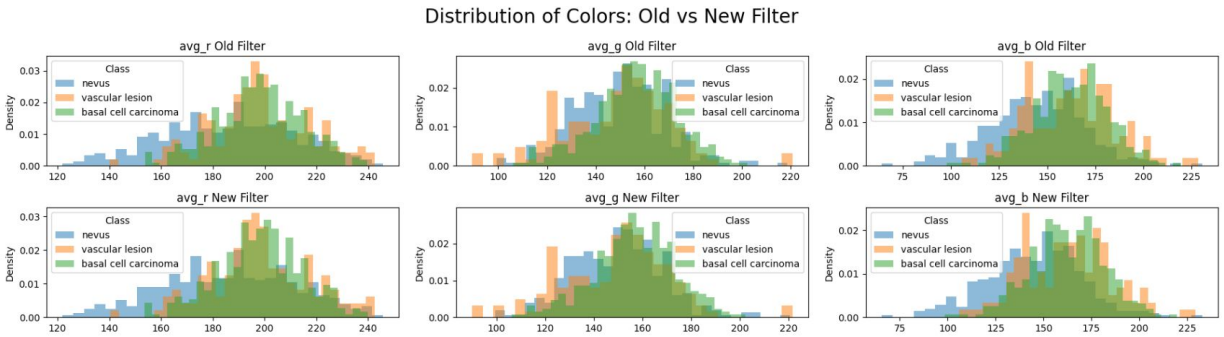


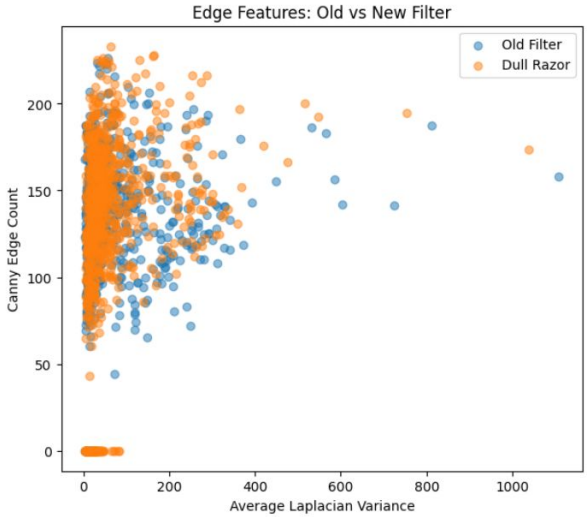
New Pipeline: DullRazor

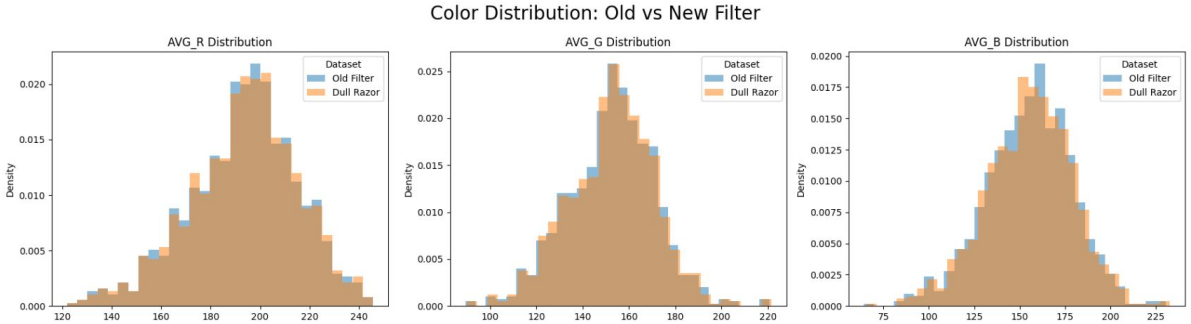
- Replaces detected hair pixels with those of the surrounding area (Lee et al., 1997)
- Significantly less number of errors
- Struggles to identify light hairs, but removes a majority of dark hairs
- New pipeline only took 3 minutes!



Analysis of New Dataset







Future Work

- Hair Filter
  - Apply new hair filter to all nine classes
- Machine Learning Models
  - Try sampling techniques (SMOTE, Random Undersampling) to handle imbalance
  - Train existing models on new filtered dataset
- CNN
  - Measure performance after hair removal using new algorithm
  - Train on an balanced dataset initially, then add more features

References

Lee, T., Ng, V., Gallagher, R., Coldman, A., & McLean, D. (1997). DullRazor: a software approach to hair removal from images. *Computers in biology and medicine*, *27*(6), 533–543. https://doi.org/10.1016/s0010-4825(97)00020-6

Velasquez, J. (2021). Dullrazor-algorithm. GitHub. https://github.com/BlueDokk/Dullrazor-algorithm